

Image Representation

Learning Objectives

- Understand how images are made of pixels
- Explain color depth and resolution
- Calculate storage needed for a simple image

Engage (5-7 minutes)

Think of a rangoli pattern made of tiny colored dots—step back and you see the full picture. Digital images work the same way—thousands of colored pixels form the complete image. Today we explore how computers store pictures.

Explore (10-12 minutes)

Open a photo on a computer and zoom in until you see individual squares—those are pixels. Count pixels along the width and height. Discuss how more pixels make clearer images. Draw a 4x4 grid and color each cell to make a simple pixel art.

Explain (10-12 minutes)

Images are grids of pixels—each pixel stores a color value. Resolution is pixel count—640x480 means 640 columns and 480 rows. Color depth determines how many colors each pixel can show—8-bit means 256 colors, 24-bit means 16 million colors. More pixels and more colors mean larger files.

Elaborate (8-10 minutes)

Calculate storage for a 100x100 pixel image with 8-bit color: $100 \times 100 \times 8 = 80,000$ bits. Convert to bytes by dividing by 8. Then calculate for 24-bit color. Discuss why high-resolution photos take more space on the phone.

Evaluate (5-8 minutes)

Exit ticket: Define resolution and color depth. Calculate the total pixels in an 800x600 image.